

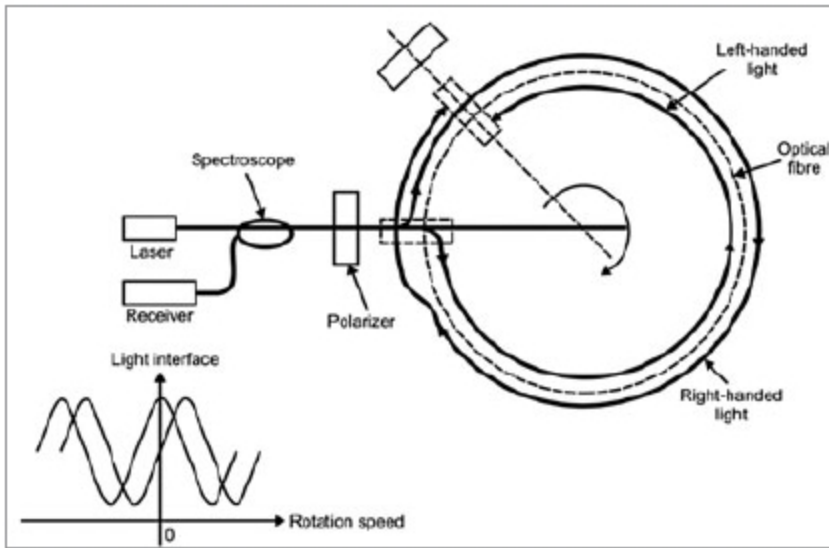


Fig. 14: Principle of operation of FOG

LN-100G is another widely-used inertial navigation system from Northrop Grumman. It is a non-dithered INS/GPS, utilising Zero-Lock gyroscope and various GPS solutions.

Fibre-optic gyroscope

Unlike a classical mechanical gyroscope based on the principle of conservation of momentum and very much like RLG, fibre-optic gyroscope (FOG) has virtually no moving parts and no inertial resistance to movement. Operational principle of FOG is also based on Sagnac effect. It consists of a long coil of optical fibre typically a few kilometres in length. It makes use of interference of light to detect mechanical rotation. FOG provides extremely precise rotational rate information, due to its lack of cross-axis sensitivity to vibration, acceleration and shock.

It does not require any starting calibration and consumes relatively less power. FOG can be very compact, as optical fibre lengths as long as a few kilometres can be wrapped around a small diameter coil.

Development of FOGs got a big boost and became a reality in the early 1970s due to technological advances in semiconductor diode lasers and low-loss single-mode optical fibre mainly triggered by the demands of the telecommunications industry.

In the case of a FOG (Fig. 14), two split-laser beams from a single laser are injected into the same fibre but in opposite directions. Due to Sagnac effect, the beam travelling against the rotation experiences a slightly shorter path length delay than the other beam. The resulting differential phase shift is measured through interferometry, thus,

translating one component of the angular velocity into a shift of the interference pattern. This is measured photometrically to derive rate information.

Beam splitting optics launch light from a laser diode into two waves propagating in clockwise and counter-clockwise directions, through a coil consisting of many turns of optical

fibre. Strength of the interference signal depends on the effective area of the closed optical path, according to Sagnac effect. This is not simply the geometric area of the loop but is enhanced by the number of turns in the coil.

FOG was first proposed by Vali and Shorthill in 1976. Development of both passive interferometer type of FOG (IFOG) and a newer concept, the passive ring resonator FOG (RFOG), are proceeding in many companies and establishments worldwide.

FOG provides extremely precise rotational rate information partly because of its lack of cross-axis sensitivity to vibration, acceleration and shock. It is also because FOG has no moving parts and does not rely on inertial resistance to movement. Hence, it is perhaps the most reliable alternative to the mechanical gyroscope.

Because of their intrinsic reliability, FOGs are used for high-performance space applications. As compared to RLGs, these typically offer a higher resolution but have relatively poorer drift and scale factor performance specifications.

Typically, FOG's sensitivity figure of $0.1^\circ/\text{hr}$ still does not match anywhere close to the demonstrated RLG sensitivity of better than $0.001^\circ/\text{hr}$. Also, unlike RLG, where zero beat frequency always means zero angular

velocity, in the case of FOG, one needs to determine which indication corresponds to zero angular velocity.

The reason for the success of FOGs lies in its desirable features like lightweight, small size, limited power consumption, projected long life time and low cost. This makes these particularly attractive in applications such as automotive and robotics. These applications call for less demanding performance levels in terms of sensitivity and drift specifications, and where small size and low cost are important considerations.

FOGs are used in a wide variety of applications. Some common ones include inertial navigation, navigation of remotely-operated vehicles (ROVs) and autonomous underwater vehicles (AUVs), surveying, automotive industry and robotics, EO/FLIR/radar stabilisation, line-of-sight tracking and precision pointing, gunfire control systems, platform stabilisation, gyroscope compassing and target acquisition systems, smart munitions and earth observation, scientific and telecommunication satellites.

Fig. 15 and 16 show two such FOGs. Fig. 15 shows EMP-1.2K from Emcore Photonics Systems. It provides precise navigation with 1 mile/hr accuracy without GPS and fast gyroscope compassing to 1 milli-radian. Major performance specifications of the device include short-term and long-term drift stability specifications of $0.005^\circ/\text{hr}$ and $0.01^\circ/\text{hr}$, respectively, scale factor stability and linearity specifications of 50PPM and 25PPM, respectively. Random walk specification is $0.0015^\circ/\sqrt{\text{hr}}$.

Fig. 16 shows 3-axis FOG assembly from AL Cielo. Sensors can be used in diverse navigation and control applications, such as for stabilisation of missiles, in airborne sensor systems or land vehicle systems. State-of-the-art FOGs compare very closely with the specifications of RLGs.

FOG-based IMUs and INSes from KVH Industries Inc., including 1725



Fig. 15: FOG type EMP-1.2K



Fig. 16: 3-axis FOG-based IMU



Fig. 17: 1725 IMU